LAYAN KAUSHIK

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EDUCATION

Duke University, Durham, NC, USA *Aug 2022 – May 2024*
M.S. Economics and Computation **GPA: 3.91/4.0**
Courses: Theory and Algorithm of Machine Learning, Probabilistic Machine Learning, Elements of Deep Learning, Design and Analysis of Algorithms, Advanced Econometrics, Numerical Analysis, Graph Network Analysis, Algorithmic Game Theory.

Indian Institute of Technology, Roorkee, India *Jul 2016 – May 2020*
B.Tech. Civil Engineering, First Division with Distinction **GPA: 8.726/10**
Ranked 12 in a class of 121 students.
Courses: Computer Programming and Numerical Analysis, Engineering Mathematics, Probability and Statistics, Data Mining, Linear Optimization, Graduate Econometrics.

HONORS AND AWARDS

Dean's Research Award for Master's Students, Duke University *2023*
U.S. Army Corps of Engineers Institute for Water Resources Research Grant, Washington DC *2023*
Duke Graduate Working Group on Global Issues Award, Duke University *2023*
Duke Economics Master's Scholar Award, Duke University *2022-24*
K. C. Mahindra Scholarship for Post Graduate Studies Abroad, K. C. Mahindra *2022*
Annual Excellence Award, IIT Roorkee Heritage Foundation *2018* & *2019*
Chief Minister's Scholarship, Government of Assam, Assam, India *2018*
IASc-INSA-NASI Summer Research Fellowship, Indian Academy of Sciences *2018*
IIT Roorkee ENCORE Award, IIT Roorkee *2017*
NEC Merit Scholarship, Directorate of Technical Education, Government of Assam *2016*
Top 3 in State Mathematics Olympiad, Homi Bhabha Centre for Science Education, India *2014*

WORK EXPERIENCE

Economist Apprentice, Amazon, Seattle, US *Jun 2024- Present*

- Working alongside Professor Alexandre Mas and Mingyu Chen at the intersection of economics and statistics—specializing in survey methodology and causal inference frameworks.

Product Data Scietist, Mastercard, India *Aug 2020 - Jun 2022*

- Worked as a Data Analyst and as a Data Lead within Mastercard's Geographic Insight platform, ensuring data accuracy and generating key geographic insights.

Field Engineer Trainee, Schlumberger (Rebranded as SLB), India *May 2019 - Jul 2019*

- Worked as a Field Engineer Trainee in the upstream Oil & Gas division.

RESEARCH PROJECTS

Applying NLP to Uncover Insights in Influencer Marketing, Guided by Prof. Tong Guo, The Fuqua School of Business, Duke University *Jan 2024 - May 2024*

- Analyzed 1.29 million ADHD-related TikTok videos using NLP and unsupervised ML, classifying each as Ad or non-Ad.
- Applied topic modeling to identify behavioral patterns in user engagement, focusing on uncovering prevalent themes and trends within ADHD-related discussions on TikTok.
- Skills: Python, Machine Learning, Natural Language Processing, Web Scrapping, Topic Modeling

Treatment Effect Estimation With Measurement Error in Covariates and Repeat Measurements, Guided by Prof. Michael Pollmann, Duke University *May 2023 - Jan 2024*

- Derived and implemented an unbiased objective method for higher-order polynomials under homoskedastic measurement error and multiple mismeasured variables from "Identification and Estimation of Polynomial Errors-in-Variables Models" by Hausman et al.

- Collaboratively developed an R package, optimizing existing codes for time and space efficiency, and executed extensive large dataset simulations on Duke's Computer Cluster.
- Implemented Kenichi Nagasawa's "Treatment Effect Estimation with Noisy Conditioning Variables" in R, demonstrating the model's versatility across various datasets.
- Crafted and executed cross-validation code in R for CoCoLasso regularization in linear regression and linear inverse probability weighted estimation, addressing datasets with measurement errors.
- Skills: R, C++, Causal Inference, Machine Learning

Data Enhancement and Machine Learning for Enhanced Flood Damage Assessment, Guided by Prof. Mark Borsuk, Duke University ***May 2023 - April 2024***

- Presented a conference poster at the American Geophysical Union (AGU) 2023 on data augmentation and feature engineering of the FEMA NFIP Version 2 Flood Insurance Claims dataset (Publication Link)
- Enhanced mapping precision using Python geospatial libraries, generating more precise new geographic shapefiles for analysis.
- Addressed dataset inconsistencies, including missing data and measurement discrepancies, to enhance the reliability of the Flood Insurance Claims dataset.
- Built outlier detection models to refine water-depth anomalies; presented findings at the AGU 2023 conference.
- Skills: Python, Geospatial Data Analysis, Feature Engineering

Spatial Distribution of Moisture Content in Cement Mortar Structures using Radial Basis Functions, Guided by Prof. Bishwajit Bhattacharjee, IIT Delhi ***May 2018 - Aug 2018***

- Conducted extensive research on structural engineering, sustainable environment, basic electrical science, and interpolation methods to measure moisture content in cement mortar required to obtain durability and service life of the structure.
- Incorporated an efficient, accurate, and reliable method to estimate the moisture content in laboratory-synthesized cement concrete blocks using data obtained from experiments conducted during the research.
- Utilized interpolation techniques like the Gaussian and inverse multiquadric radial basis function method in the estimation function to determine moisture content for which data could not be generated and verified with the theoretical calculations.
- Skills: Python (Libraries: NumPy, Matplotlib, SciPy)

ACADEMIC PROJECTS

Strategy-Proof Tournament Design for Multi-Stage Tournaments, Guided by Prof. Kamesh Munagala, Duke University ***Feb 2024 - April 2024***

- Investigated incentive compatibility in multi-stage tournaments, where players may strategically underperform to secure easier future matchups (e.g., drop-down tournaments in esports and university competitions).
- Proposed a strategy-proof reward system using a scaled bracket-based incentive structure and a dynamic Elo-style rating update; proved incentive compatibility mathematically and validated the model through Monte Carlo simulations across varied player strategies.
- Applied theoretical insights to real-world tournament formats (e.g., Pickleball, UEFA Champions League), revealing design flaws and suggesting incentive-aligned alternatives.
- Skills: Mechanism Design, Monte Carlo Simulation

Addressing User Activity Bias in Bipartite Graph Ranking, Guided by Prof. Xiaobai Sun, Duke University ***Sep 2023 - Dec 2023***

- Enhanced the BiRank algorithm for bipartite graph rankings in recommendation systems, incorporating user activity bias mitigation through a novel regularization technique.
- Developed a unique fairness metric, combining activity rank balance and Gini coefficient, to evaluate the improved algorithm's effectiveness in diverse graph scenarios.
- Executed extensive experimental analysis on synthetic datasets, demonstrating significant fairness improvements in rankings, particularly for power-law distributed graphs.
- Skills: Advanced algorithm design, Experimental Design

Ride Sharing and Tipping Behavior in Chicago from 2018 to 2023, Guided by Prof. Jian Pei, Duke University *Jan 2023 - May 2023*

- Analyzed 300M+ Chicago ride-sharing records (2018-2023) using regression, time-series analysis, and hypothesis tests; identified statistically significant patterns, including a 33% ride drop post-pandemic and 1.5% tipping increment, reflecting socio-economic trends.
- Utilized Python for large dataset management; implemented chunk-based data processing, batch processing, and dynamic sampling to optimize data tasks.
- Skills: Time-Series Analysis, Regression, Hypothesis Testing

Uber Movement Data replication and Here Maps API simulations, Guided by Prof. Amit Agarwal, IIT Roorkee *Jan 2019 - May 2019*

- Extracted real-time data from HERE MAPS API and successfully compared it with the performance indicators of Uber Movement data of New Delhi to detect patterns and analyze the impact of major events like national holidays, festivals, and rush hours on traffic conditions.
- Replicated the indicators of Uber Movement data using open-sourced data.
- Skills: Python (Libraries: osmnx, pandas, NumPy, Matplotlib, geopandas)

TEACHING EXPERIENCE

Graduate TA, CS230: Discrete Math for CS, Duke University *Jun 2023 - Aug 2023*
Responsibilities include conducting office hours, developing recitation slides, and delivering lectures.

Graduate TA, CS216: Everything Data, Duke University *Jan 2023 - May 2023*
Responsibilities include holding office hours, managing undergraduate TAs, creating assignments, and automating grading.

Undergraduate TA, Computer Programming in C++, IIT Roorkee *Jan 2018- May 2018*
Held office hours and lab sessions with over 40 students.

Undergraduate TA, Solid Mechanics, IIT Roorkee *Jul 2017 - Nov 2017*
Held office hours and tutorial sessions with over 40 students.

SKILLS

Programming Languages: Python (scikit-learn, PyTorch, pandas, statsmodels, econml, geopandas), SQL, R, C++
ML Frameworks: PyTorch, JAX
Software Packages: Stata, Tableau, MS Excel, LaTeX
Spoken Languages: English (fluent), Assamese (native speaker), Hindi (fluent)

VOLUNTEER WORK AND OTHER ACTIVITIES

Assistant Badminton Coach, Duke University *Aug 2022 - May 2023*
Provide regular training and coaching advice to players affiliated with Duke Badminton Club.

Student Mentor, Student Mentorship Program, IIT Roorkee *Jul 2018 - May 2019*
Mentored a batch of 7 freshmen to help with their transition from high school to university.

Professional Player Badminton, India *Jan 2010 - May 2020*
Represented the Indian state of Assam in various Sub-Junior and Junior National Championships in India. Was awarded Best Badminton Player of IIT Roorkee for three consecutive years 2017, 2018, and 2019.